Attention Residuals

Deep Dive into Kimi K3 — Chapter 7

Yin Yang

Foundation Books Project

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Goal: type depth mixing, define Full and Block AttnRes, place them in K3's 93-layer graph, trace a toy forward, execute it exactly, and keep the evidence in scope.

Roadmap

Depth as a source axis

Full versus Block

Execution and evidence

Depth as a source axis

A learned softmax over earlier outputs

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- Normalize keys to score, mix raw values — a probability check alone cannot catch the reverse.

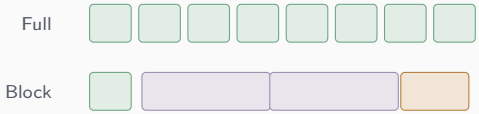
Full versus Block

Compression trades addressability for cost



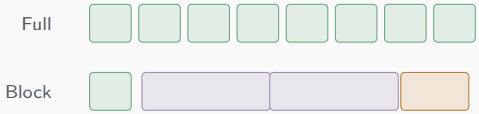
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- Source conservation: summaries and partial cover each output exactly once; commit before the next block's first target.

Execution and evidence

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Depth state lives for one forward. Next: Chapter 8's Stable LatentMoE.